```
#Q1
from sklearn.datasets import load_diabetes
import pandas as pd
import statsmodels.api as sm

diabetes = load_diabetes(as_frame=True)
df = diabetes.frame

print(df.head())
# Correlation
print(df[["bmi", "target"]].corr())

# Simple linear regression
X = sm.add_constant(df["bmi"])
y = df["target"]

model = sm.OLS(y, X).fit()

print(model.summary())
```

```
        age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6  target
0 -0.002592  0.019907 -0.017646   151.0
1 -0.039493 -0.068332 -0.092204    75.0
2 -0.002592  0.002861 -0.025930   141.0
3  0.034309  0.022688 -0.009362   206.0
4 -0.002592 -0.031988 -0.046641   135.0
            bmi   target
bmi     1.00000  0.58645
target  0.58645  1.00000
                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.344
Model:                            OLS   Adj. R-squared:                  0.342
Method:                 Least Squares   F-statistic:                     230.7
Date:                Thu, 08 Oct 2026   Prob (F-statistic):           3.47e-42
Time:                        08:08:01   Log-Likelihood:                -2454.0
No. Observations:                 442   AIC:                             4912.
Df Residuals:                     440   BIC:                             4920.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        152.1335      2.974     51.162      0.000     146.289     157.978
bmi          949.4353     62.515     15.187      0.000     826.570    1072.301
==============================================================================
Omnibus:                       11.674   Durbin-Watson:                   1.848
Prob(Omnibus):                  0.003   Jarque-Bera (JB):                7.310
Skew:                           0.156   Prob(JB):                       0.0259
Kurtosis:                       2.453   Cond. No.                         21.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Q1 Ans:

Pearson Correlation: r = 0.5865

There is a moderate positive relationship between BMI and disease progression.

Regression Equation: Y = 152.1335 + 949.4353X

$R^2$ = 0.3439

Therefore, BMI explains 34.39% of the variation in disease progression.

```
#Q2
X = df[["bmi", "bp", "age"]]
X = sm.add_constant(X)
```

```
model = sm.OLS(df["target"], X).fit()

print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.396
Model:                            OLS   Adj. R-squared:                  0.392
Method:                 Least Squares   F-statistic:                     95.81
Date:                Thu, 08 Oct 2026   Prob (F-statistic):           1.09e-47
Time:                        08:09:16   Log-Likelihood:                -2435.7
No. Observations:                 442   AIC:                             4879.
Df Residuals:                     438   BIC:                             4896.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const        152.1335      2.859     53.211      0.000     146.514     157.753
bmi          788.7827     65.562     12.031      0.000     659.927     917.638
bp           394.1271     68.392      5.763      0.000     259.710     528.544
age           25.9904     63.923      0.407      0.685     -99.643     151.623
==============================================================================
Omnibus:                       12.586   Durbin-Watson:                   1.928
Prob(Omnibus):                  0.002   Jarque-Bera (JB):                7.934
Skew:                           0.175   Prob(JB):                       0.0189
Kurtosis:                       2.444   Cond. No.                         28.0
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Q2 Ans:

$R^2$ = 0.396

Adjusted $R^2$ = 0.392

Significant Predictors: BMI ⟶ Significant (p < 0.001) BP ⟶ Significant (p < 0.001) Age ⟶ Not Significant (p = 0.685)

Regression Equation:

Y = 152.1335 + 788.7827(BMI) + 394.1271(BP) + 25.9904(Age)

Conclusion:

BMI and BP are significant predictors of disease progression, while Age is not significant.

```
#Q3
from scipy.stats import spearmanr

study_time = [2, 4, 1, 5, 3]
exam_score = [50, 65, 45, 80, 60]

correlation, p_value = spearmanr(study_time, exam_score)

print("Spearman's correlation coefficient:", correlation)
print("p-value:", p_value)
```

```
Spearman's correlation coefficient: 0.9999999999999999
p-value: 1.4042654220543672e-24
```

Ans: Study Time & Exam score are perfectively positively corelated.

```
Start coding or generate with AI.
```